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Scientific Report

Aspect-Based Sentiment Analysis for Romanian Reviews
Artificial Intelligence

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1 INTRODUCTION

Aspect-based sentiment analysis(ABSA) is as a type of text analysis that categorizes fine-grained option polarity by aspect (term or category) and identifies the sentiment attribute related to each aspect, the sentiment can have three polarities: positive, negative, or neutral. One known challenge is identifying the correct aspect, the correct polarity since the sarcasm is rarely understood.

ABSA is an advanced text analysis technique. This technique is also known as aspect-level sentiment analysis, feature-based sentiment analysis or as aspect sentiment analysis. ABSA is so important nowadays because is useful in companies to automatically sort and analyse customer data, some of the aspects that makes ABSA popular is that ABSA is scalable, performs real-time analysis, get considerate criteria, and understand people texts (e.g., customer review, twitter post) on a deeper level. A challenge of ABSA is the domain sensitivity of the opinion terms and the aspect terms. ABSA has two subtasks: Aspect-Term Sentiment Analysis (ATSA) and Aspect-Category Sentiment Analysis(ACSA).

There are many use cases where ABSA can be more useful, such as: review analyser tool that helps customers to choose their perfect product based on the desired aspects, chatbots / virtual assistants that provides personalized responses based on customer sentiments for specific aspects, employee feedback analysis that helps companies improve strategies to meet employee needs, customer feedback analysis to be implement in call centres to analyse feelings for specific aspects, healthcare feedback analysis to improve the hospitality by considering patient feedback, including aspects such as treatment, service, and price, marketing tool that analyse public reactions when launching a new product, food applications analysis that simplifies the customer experience by considering aspects such as taste and delivery time before ordering.

Previous Work.

The scope was to implement and analyse various neural network architectures for ABSA on the Multi-Aspect Multi-Sentiment (MAMS) dataset [Jiang et al., 2019]. The objective was to classify sentiments (positive, negative, or neutral) directed at specific aspects within a given text.

Data Preprocessing and Encoding for MAMS dataset was converted from its native XML format into a structured CSV file. Two primary text encoding methods were employed: One-Hot Encoding (text was converted into numerical vectors based on a pre-established vocabulary. Padding was applied to standardize sentence lengths for network input) and Transformer Embeddings (pretrained models (XLNet [Yang et al., 2019], BERT [Sun et al.,

2019], etc.) were used to generate numerical vectors, with input dictionaries containing *input_ids*, *token_type_ids*, and *attention_mask* [Chowdhury et al., 2024])

For the model architectures we developed and trained the following neural networks: text CNN, text BiGRU, hybrid model architectures combining BiGRU and CNN layers [O’shea and Nash, 2015], pretrained models (XLNet, BERT, DeBERTa, ReviewBERT) were fine-tuned for the ABSA task, hybrid transformer-BiGRU (BiGRU layers were applied over the embeddings generated by a pretrained XLNet model).

For training and evaluation the models were trained using the Adam optimizer with a categorical cross-entropy loss function, early stopping was implemented based on validation loss to prevent unnecessary epochs and restore optimal model weights, performance was evaluated using F1-Score, accuracy, precision, and recall, with detailed logs maintained for each experiment.

Our results were better by using a textCNN than the results presented in the paper released for the MAMS for ABSA dataset, and the results for BiGRU implementations were similar with the textCNN but the results for the Transformers architectures were worse.

Motivation.

While researching for a dataset in the Romanian language, we observed that there are only a few options in this direction. One example is the review dataset by [Briciu et al., 2024], which contains 12 product categories but is limited to a single aspect per review. Another is the e-commerce platform review dataset by [Tache et al., 2021], containing 15,000 reviews divided equally, half being negative and the other half positive.

The research shows that aspect-based sentiment analysis for the Romanian language currently lacks extensive exploration, making it an area with significant research potential. Even though it is a promising research area, it comes with many challenges by default, such as limited resources, often restricted to binary classification.

After we analysed the MAMS dataset and the resources available in Romanian, we were determined to include ABSA in the Romanian core. Our dataset is among the first, to our knowledge, to include mixed aspects multiple sentiments ABSA for reviews in Romanian language. Being mixed aspects represent that the dataset is an hybrid, including both single aspects reviews and multi aspects ones. Our dataset is based on e-commerce reviews for phones, covering multiple categories (e.g., low-end, mid-range, high-end, traditional phones with buttons), making it versatile. Although the dataset is unbalanced, it accurately represents the real world, where people may be succinct about their experiences. Considering that customer service in Romania and worldwide is constantly evolving, building a dataset where Romanian users can provide valuable insights is highly relevant.

Objectives.

The main objectives are the following.

- *Data acquisition and cleaning: Gather, analyse and preprocess the data for the Romanian*

Review Dataset for phone reviews, a comprehensive dataset suitable for ABSA tasks.

- *Ground truth annotation: Develop and deploy an annotation tool to facilitate manual labelling of the test dataset, crucial for model evaluation.*
- *Automated annotation with LLMs: Use LLMs with few-shot example learning to generate training data annotations.*
- *Knowledge distillation for small transformers: Leverage the generated training data to train a smaller, specialized student model optimized for production deployment.*

Summary of the Scientific Report.

This Scientific Report is structured into **6 chapters**, each of which plays an important role in developing a comprehensive understanding of the research subject, related works, methodology, implementation, experimental results, and conclusions. Chapter 2 presents a survey of articles and analyses state-of-the-art models that can be applied, with some studies being used to improve our models and gain deeper knowledge about the subject. Chapter 3 provides an overview of the dataset we created and used, the architecture of the employed models, and details about the implementation. Chapter 4 offers a detailed step-by-step walk-through; Section 3.1 describes the process of creating the Mixed-Aspects Multiple-Sentiments Romanian Review (RoR) for ABSA dataset, from data gathering to annotation. The remainder of the chapter discusses the experimental setup, results for each model, and observations regarding challenges encountered. Chapter 5 focuses on the performance of our models on the RoR dataset, while Chapter 6 highlights conclusions, future directions, and potential solutions to the challenges identified.

2 RELATED WORK

In this section are analysed inspirational papers that are part of our process of developing.

Dumitrescu et al. [2020] highlight the unique contributions of their study, introducing the Romanian BERT. Their model is built upon the foundation of transformer-based architectures, which have revolutionized the field of NLP. Since the introduction of the Transformer model, a number of significant pretrained language models have emerged. These models, such as BERT [Devlin et al., 2018], have demonstrated remarkable performance across a wide range of tasks. The authors have trained their model on various Romanian datasets, leaving behind an explicit guideline to help the community replicate their work.

The 3 corpus they used are publicly available corpora, OPUS [Tiedemann, 2012] that represents a collection of translated text, it is interesting that the gathering of the corpus, is entirely automated, resulting in 4GB of Romanian text, followed by Open Super-large Crawled ALMAAnaCH coRpus(OSCAR) [Suárez et al., 2019] that represent a large multilingual corpus, the Romanian section containing 11GB, and the last one being Wikipedia, specifically the Romanian Wikipedia that is publicly available, they took only 0.4GB of text after cleaning it, since the Wikipedia needed extra cleaning procedures.

The evaluation of the proposed model was made on 3 downstream tasks, each task was compared with M-Bert, including both cased and uncased versions. Simple Universal Dependencies, Named Entity Recognition, both had overperformed M-BERT on all subtasks, for the first one, the uncased Romanian BERT model performed better, on the other hand, for the second, the cased Romanian BERT performed better, on the other hand, for Joint Universal Dependencies, the results were mixed, only Lemmas performed better on M-BERT (cased), for UPOS Romanian BERT (uncased), and Romanian BERT (cased) performed better on both UFeats and LAS.

In conclusion, the authors introduced the first BERT model, that was pretrained on approximately on 15BG of Romanian text, they are looking forward to introduce new versions of BERT.

Masala et al. [2020] present Romanian-only pretrained BERT(RoBERT - small, medium and large), and they want to test it, on different Romanian specific NLP tasks, RoBERT being compared with multi-lingual and mono-lingual for Romanian BERT models, resulting in RoBERT keep outperforming them.

The corpus was extracted from various sources, the main ones being OSCAR and Romanian Wikipedia dump, gathering 12.6GB of data.

The architecture is similar to BERT, being based on a multi-layer bidirectional Transformer, even the models RoBERT-base and RoBERT-large having the same layer sizes as BERT-base and BERT-large.

The models were trained in the same manner as BERT, using masked language modeling and next sentence prediction objectives. The Adam [Tiedemann, 2012] optimizer was used with BERT hyperparameters. The tokenization process was adapted to handle Romanian diacritics.

Since there are no large benchmark collection for Romanian, the models were evaluated on seven downstream tasks across 3 categories: sentiment analysis, dialect tasks and automated diacritics restoration.

The experiments followed a standard training procedure: models were trained on the training sets for multiple epochs, after that, only the best performing model on the dev set was selected than evaluate it on the test set.

The paper concludes after the experiments that the mono-lingual models have better results than multi-lingual ones, even though it had less resources, but this challenge only accentuated the struggles of having limited resources in countries similar with Romania.

Jiang et al. [2019] introduce a new dataset for Aspect-Based Sentiment Analysis in which each sentence contains at least two aspects with different sentiments. To ensure accuracy, the data was manually annotated, resulting in a large-scale multi-aspect, multi-sentiment resource that enables more advanced ABSA research and outperforms existing state-of-the-art datasets.

For the data collection the approach was similar to the SemEval-2014 Restaurant Review dataset, the sentences of the Citysearch New York dataset were annotated and each document was split to form a corpus, then sentences were removed in which there were more than 70 words.

Dataset		Pos.	Neg.	Neu.	Total.
ATSA	Train	3380	2764	5042	11186
	Train (small)	1089	892	1627	3608
	Validation	403	325	604	1332
	Test	400	329	607	1336
ACSA	Train.	1929	2084	3077	7090
	Train (small)	1000	1100	1613	3713
	Validation	241	259	388	888
	Test	245	263	393	901

Figure 1: Statistic for MAMS dataset [Jiang et al., 2019]

For the annotation data, two versions of the MAMS dataset were created for two subtask sites, these being: aspect-term sentiment analysis (ATSA) and aspect-category sentiment analysis (ACSA). For dataset analysis, the statistics results are represented in Figure 1.

The proposed model contains four layers: embedding, encoding, primary capsule and category. The layers are represented in Figure 2.

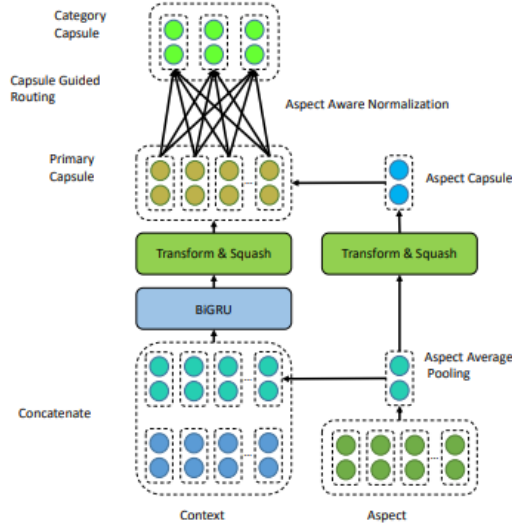


Figure 2: The layers of the proposed model [Jiang et al., 2019]

The article introduces an advanced ABSA dataset and shows strong potential for transformer- and BERT-based methods.

For experiences, testing was done using 300-dimensional word vectors pre-trained by GloVe [Pennington et al., 2014], Adam optimizer [Kingma and Ba, 2017], [Kamila et al., 2022] was used for train, with a learning rate of 0.0003 and 0.00003 for CapsNet [Sabour et al., 2017] and CapsNet-BERT the models were run 5 times, and as a weak point following the results it is noticed that MAMS dataset has worse results on certain tasks than state-of-the-art, an example would be sentence-level sentiment classifiers and attention-based models without effectively modelling word order.

Alqaryouti et al. [2024] present a model that helps government agencies to understand better the customer needs by analysing application reviews thus improving the service delivery. The model employs a hybrid approach to aspect-based sentiment analysis that uses sentiment extraction techniques using lexicons and rule-based methods, alongside aspect extraction to identify targeted opinions and highlight issues in government mobile apps.

Other related work underlines that sentiment representation involves five core elements such as, entity, target, sentiment, opinion holder and timestamp which collectively capture the essential aspects of a sentiment expression. They imply that sentiment can be analysed at different levels such as document-wide, sentence-level or aspect specific which allows for varying degrees of granularity in understanding the sentiment. Aspect-based classification plays an important role by breaking down entities into more specific aspects to better capture nuanced sentiments, while concept-level sentiment analysis focuses on semantic understanding but faces limitations in practical applications. They say that there has been explored many traditional approaches for aspect extraction such as POS tagging and machine learning techniques such as SVM classifiers but recent studies have also investigated deep learning and unsupervised techniques

to improve sentiment analysis. However, challenges remain, particularly the need for labelled datasets in the new domains thus limiting the opportunities of deep learning.

The authors build on the dataset proposed by [Alqaryouti et al., 2018], which covers domain-specific government smart applications enriched with lexical resources from [Alqaryouti et al., 2024]. The dataset was manually annotated and verified, consisting of short reviews that often include slang, spelling errors, informal grammar, and occasional ineffective emoticons. Given these characteristics, the authors adopted an integrated lexicon and rule-based model. The dataset was split into a training set of 5,141 reviews (70%) and a test set of 2,205 reviews (30%). Evaluation relied on confusion matrices and standard metrics such as precision, recall, F1-score, and accuracy, as well as a personal correlation coefficient to measure how closely predicted sentiments aligned with the gold standard (values near 1 indicating strong correlation).

For preprocessing, reviews were segmented into sentences using punctuation markers such as periods, question marks, and exclamation points. The review subject was prepended as the first sentence, while punctuation and stop words (based on a custom list) were removed, and all text was lowercased. Importantly, repeated characters were retained to preserve semantic meaning (e.g., "greaaaaat" signaling a very positive sentiment).

In the context of aspect extraction this paper uses both implicit and explicit aspects, categorizing them into User Interface, User Experience, Functionality and Performance. Explicit aspects are directly mentioned in the text while the implicit aspects are inferred from opinion words. Their algorithm identifies these opinion words and maps them to the nearest relevant aspect terms.

For sentiment scoring, they used lexicons (i.e., pre-made lists) to analyse sentences for opinions about specific aspects like a product feature. When the algorithm finds an opinion word, it looks up to its polarity score which indicates whether the opinion is positive or negative. The algorithm also implies various rules to adjust the scores based on factors like negation (e.g., "not attractive") and intensity (e.g., "very good"), ultimately returning two lists, one for polarity scores and another one for the corresponding aspect categories.

The algorithm is designed to analyse user reviews and assign star ratings to different aspects of an application based on the sentiments expressed in the reviews. It uses a five-star scale, where one star indicates a very negative opinion and five stars indicate a very positive opinion. By calculating average sentiment scores for specific aspects, the algorithm helps application developers understand what users like or dislike, allowing them to improve their products based on user feedback.

In conclusion, the integrated lexicon and rule-based approach enabled effective detection of both explicit and implicit aspects, improving classification performance.

Jin et al. [2024] propose a model called Target Word Transferred ABSA (WordTransABSA), which leverages the pre-training task of the Masked Language Model (MLM) [Jin et al., 2024] to enhance aspect-based sentiment analysis. By transferring output tokens corresponding to

target words to predefined sentiment-related pivot tokens, WordTransABSA aims to predict sentiment polarities effectively, akin to the MLM’s approach of predicting masked tokens. This model diverges from [Jin et al., 2023] traditional fine-tuning methods by using the entire parameter set of pre-trained language models, including both encoder and decoder components, thereby maximizing the exploration of prior knowledge embedded in pre-trained language models. This model has three modules such as the identification of sentiment-related pivot tokens, the generation of probabilities for these tokens and the mapping of sentiment words. It uses BERT for encoding and decoding input sequences, allowing for the effective generation of targeted sentiment sequences based on predefined pivot words. Experimental results indicate that it outperforms traditional pre-trained language model fine-tuning methods [Jin et al., 2024], particularly on low resource scenarios.

The sentiment pivot token engineering proposed focuses on sentiment-related words for a better word substitution. This process uses domain expertise for selection strategies together with prompt learning methodologies to improve the model performance. The document retrieval part for this emphasized the use of the Term Frequency-Inverse Document Frequency (TF-IDF) technique to identify and rank sentiment-related words, this process involves filtering and selecting high-quality sentiment indicators based on their contextual relevance and classification capability, therefore facilitating more accurate sentiment analysis in NLP tasks. They discuss using two prominent sentiment lexicons such as NRC Word-Emotion Association Lexicon and OpenHowNet to systematically identify and filter sentiment-related words based on their emotional associations. The NRC Emotion Lexicon provides a structured list of words linked to both coarse and fine-grained sentiments, while OpenHowNet [Qi et al., 2019] serves as a filtering tool to refine the output from NRC by eliminating noise words, ultimately yielding a robust set of candidate sentiment pivot words for further analysis. Then they integrate the discovered sentiment-related words by using weight-based operations, where the intersection of words from both document retrieval and sentiment lexicon searches is selected for positive and negative categories.

The WordTrans LM leverages sentiment-related pivot tokens during fine-tuning. Input sentences are first encoded into low-dimensional vectors using BERT, where the traditional MLM task is adapted to predict sentiment-specific tokens at predefined positions, preserving BERT’s pretraining strengths. Effective word integration, achieved through strategies such as average pooling and K-Medoids clustering, ensures robust sentiment classification with computational efficiency.

To resolve gap between BERT’s output logits (word probability distributions) and fine-grained sentiment polarity, WordTrans LM introduces a two-dimensional vector representation conditioned on sentence length. Two detection strategies are applied: Horizontal Detection, which selects the highest-probability word at the aspect term position, and Vertical Detection, which re-ranks sentiment-related pivot tokens if the first strategy fails. This pipeline enhances both efficiency and robustness in sentiment analysis, while keeping the underlying language model architecture intact and fully exploiting pretrained knowledge.

3 METHODOLOGY

This chapter provides a brief introduction to the overall architecture, the models used, and the Mixed-Aspects Multiple-Sentiments Romanian Review (RoR) for ABSA dataset. To ensure the overall quality of our models the testing data was manually annotated, while large language models (GPT-3.5, GPT-3.5-turbo [OpenAI, 2023], and GPT-4.1 [OpenAI, 2025]) [Radford et al., 2018] were used for automatic train labeling. For model training, we used BERT Romanian cased V1 [Dumitrescu et al., 2020] and RoBERT [Masala et al., 2020] as baselines.

3.1 RoR Dataset

Reviews were sourced from a well known e-commerce platform, eMAG [eMAG, 2025], identified as the most representative Romanian electronics platform due to its widespread consumer use. To overcome restrictions that prevented direct access to review pages via Python's request library (i.e., not being able to acquire a new review page from the URL), a Selenium-based scraper was developed to automate navigation across paginated content. Review links were selected manually and stored in a structured JSON file, which allowed for semi-automated batch processing while respecting platform request limits. We gathered a total of 120 mobile phone links, each with a minimum threshold of 50 reviews and assigned unique identifiers. This procedure yielded a corpus of 38,953 reviews, from which relevant review-level data such as, review title, text body, number of likes and rating given, was systematically extracted.

We scraped Wikționar (Romanian Wiktionary) [Wiktionary contributors, 2025] using Selenium, supplemented it with data from an additional online dictionary, and merged the two sources into one comprehensive dictionary. The data were normalized by replacing diacritics with their international equivalents, then consolidated into a final dictionary with duplicates removed.

Our analysis focused on 14 aspects such as: experience, price-quality, durability, camera, audio, battery, brand, design, quality, performance, accessories, compatibility, software and service. Each sentiment was categorized according to three polarities: Positive, Negative and, Neutral.

3.2 Architecture

We present the overall architecture in Figure 3. The figure illustrates a teacher–student pipeline for ABSA, where we give as input a prompt containing instructions and few-shot examples [Brown et al., 2020] of annotated reviews together with a batch of review data to a Large Language Model (LLM). The LLM serves as the teacher for a lightweight Transformer [Vaswani

et al., 2017] model acts as the student. The teacher LLM will provide automatically annotated train data for our RoR model which will learn to predict aspects and their respective sentiment by using a custom loss function.

A customer review, identified by an ID, Title, and Review, is processed as the first input. The example shown includes a Romanian review text, **"E excelent ma tine de cand l-am cumparat ii tine bateria mult, cularea e foarte frumoasa"**, which translates to positive feedback about product durability, battery life, and design aesthetics.

The LLM, guided by instructions and few-shot learning, first extracts candidate aspects (e.g., Experience, Durability, Battery, Design) and assigns sentiment polarities.

The RoR Transformer Model (student) then learns to refine and generalize these predictions, enabling efficient deployment with reduced computational overhead compared to the LLM. The final structured output is a set of aspect–sentiment pairs.

The joint optimization is performed with a composite loss function:

$$\mathcal{Loss} = \alpha \cdot \text{BCE}(\hat{y}_{\text{aspect}}, y_{\text{aspect}}) + \beta \cdot \text{CE}(\hat{y}_{\text{sentiment}}, y_{\text{sentiment}}, w_{\text{sentiment}}) \quad (1)$$

where BCE (Binary Cross-Entropy) ensures accurate multi-label aspect detection, and CE (Cross-Entropy) enforces correct sentiment classification. The coefficients α and β represent weights for each loss to encourage the aspect detection or sentiment classification more during training. The sentiment class weights $w_{\text{sentiment}}$ is optional and was used to mitigate imbalance among sentiment classes.

This teacher–student architecture helps the student model to inherit the LLM's reasoning ability while maintaining efficiency, combining the strengths and generality of an LLM and the inference performance and compactness of a smaller, more focused transformer model.

In Figure 3 the system processes review inputs for two models: a Large Language Model with instructions and few-shot examples, and our RoR Transformer Model. Both models extract sentiment predictions for multiple aspects mentioned in Section 3.1. The RoR model is trained using a combined loss function of Binary Cross-Entropy for aspect detection and Cross-Entropy for sentiment classification.

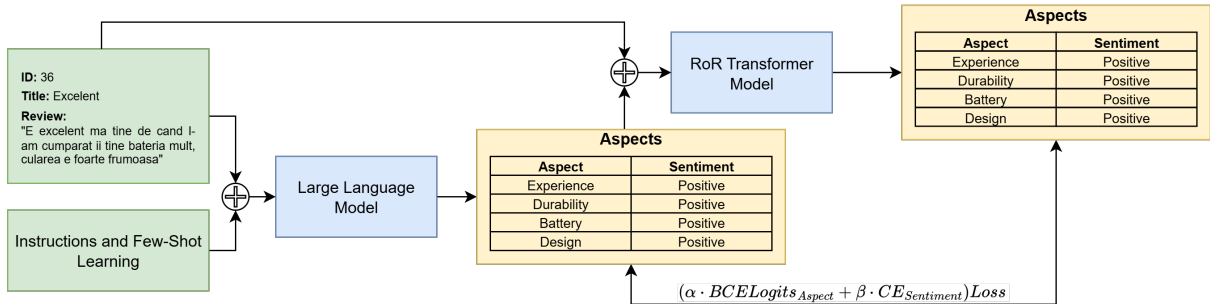


Figure 3: Overall RoR (Romanian Review) architecture for aspect-based sentiment analysis.

4 IMPLEMENTATION

In this chapter, we discuss the whole process from gathering information to actually create a dataset, our dataset being one of the few in the world to be able to handle both single aspects and multiple aspects per review and multiple sentiments, on Romanian language being less resources for ABSA than the English language. To demonstrate the efficiency of the dataset, we prepared some experiments, some of them are explained in more detail in chapter 5.

4.1 Dataset - Romanian Review(RoR) for Aspect-Based Sentiment Analysis Dataset

This section presents the Romanian Review (RoR) dataset, a novel corpus of over 30,000 Romanian product reviews collected from e-commerce platforms with both manually and automatic annotation for aspect-based sentiment analysis. The dataset creation process involves data collection, linguistic analysis, manual and automated annotation across 14 product aspects, and comprehensive preprocessing to produce a mixed-aspect, multi-sentiment dataset that addresses the lacking of Romanian ABSA datasets.

4.1.1 Pervious Work and Inspiration

The Multi-Aspect Multi-Sentiment (MAMS) [Jiang et al., 2019] dataset for ABSA dataset was an inspiration for our dataset, it has two directories, one that represents aspect-term sentiment analysis (ATSA) and aspect-category sentiment analysis (ACSA) which has the data already splitted into train, test and validation sets. It also has a JSON file that contains all the polarities and some words referring to that specific polarity (e.g., outstanding for a positive polarity).

For the JSON file, there are 3 polarities, positive, negative and neutral, each having exactly 100 words. In the figure below we can observe the balance of the words chosen by the dataset authors for each type of polarity.

Then, it was studied the influence of categories and of the terms for both versions of the dataset, ACSA and ATSA respectively. There were seven categories such as, menu, ambience, miscellaneous, staff, service, food, price and place, those were for train, test and validation sets. For the terms version of the dataset the situation was different, there were 2586 unique terms in the train set, 519 unique into the validation set and 531 unique into the test set, as a

MAMS Dataset	ACSA			ATSA		
	Train	Validation	Test	Train	Validation	Test
Total Sentences	3149	400	400	4297	500	500
Longest Sentence	67	58	64	67	59	64
Shortest Sentence	4	5	5	4	5	5
Total unique Words	10458	2558	2640	13225	3249	3272
Negative Polarity	2084	259	263	2764	325	329
Neutral Polarity	3077	388	393	5042	604	607
Positive Polarity	1929	241	245	3380	403	400

Table 1: Data analysis of the original MAMS dataset.

further analysis the number of terms that appear in both train and validation are 281, the number of terms that appear in both train and test are 266 and the number of terms in both validation and test are 170.

In order to understand better our data and to be able to train it at the most optimal space and time complexity other values were studied. Firstly, the total number of texts for each review in each train, validation and test subset of each dataset version, then the longest and shortest sentences in each dataset in order to find out the optimal padding for the network, after that the total of unique words in each dataset, in order to generate a vocabulary that can be used for diverse embeddings.

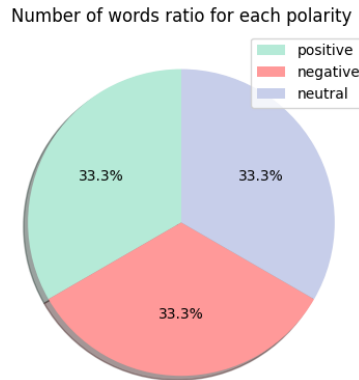


Figure 4: Sentiment polarities of MAMS overall dataset with train, validation and test subsets combined.

After the raw data analysis, in order to train the network, it was chosen to create another variant of the dataset as CSV, where the header had titles for the columns as text, category and polarity for the category version of the dataset and for the term version it had text, term, polarity, begin, end. The CSV variant of the dataset has each sentence with each polarity.

So, as can be observed in the Table 3, in the dataset a single propriety has changed by

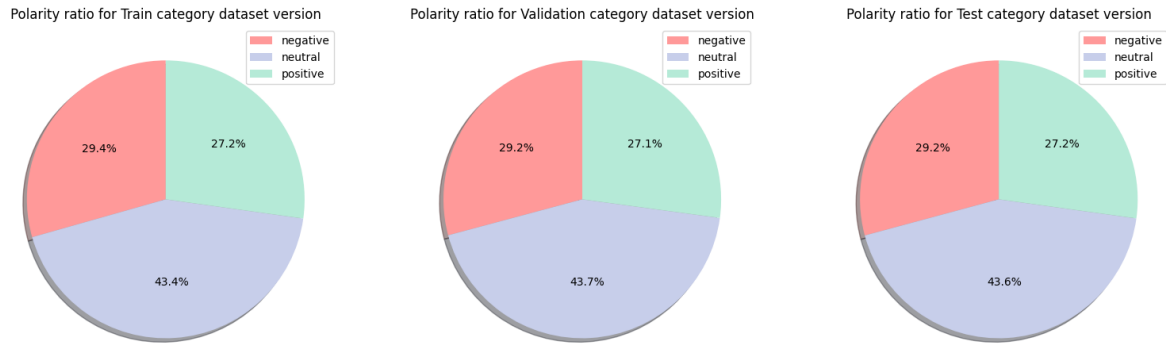


Figure 5: Sentiment polarities Ratio of MAMS dataset for aspect-category sentiment analysis. On the left are the sentiment polarities for the train set, for validation in the middle and for test on the right.

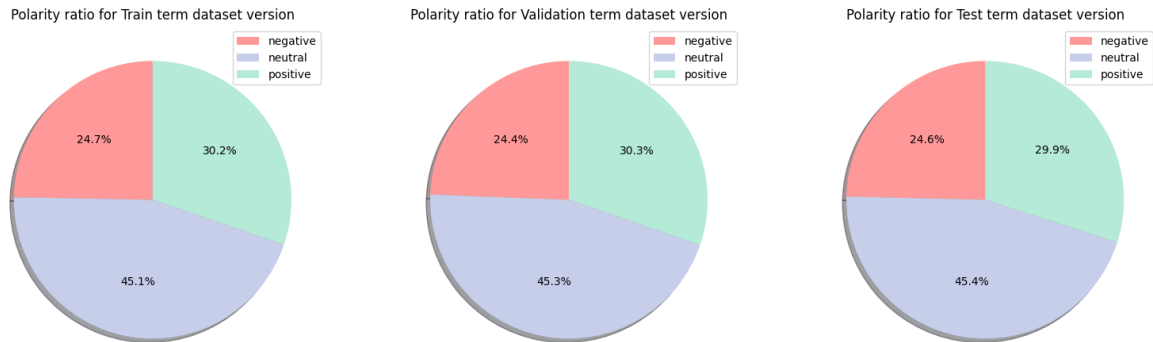


Figure 6: Sentiment polarities Ratio of MAMS dataset for aspect-term sentiment analysis. On the left are the sentiment polarities for the train set, for validation in the middle and for test on the right.

transforming from JSON file to CSV format, the number of sentences.

4.1.2 Gather dataset

The reviews were gathered from the eMAG e-commerce platform [eMAG, 2025]. After analyzing each major Romanian electronics site, we concluded that eMAG is the most suitable platform for review collection, as it is the one of the most widely used in Romania for online electronics purchases and contained the largest volume of reviews, facilitating our data scraping process.

Initially, a simple scraping script was created to obtain the data. However, the reviews are paginated in batches of 10 reviews per page, and obtaining the next batch of reviews required clicking the "next page" button, which was not possible within the simple Python requests package. To enable navigation through pagination, we built a scraper using Selenium that opened a browser instance and automated the button clicks necessary to gather the complete

MAMS Dataset	CSV ACSA			CSV ATSA		
	Train	Validation	Test	Train	Validation	Test
Total Sentences	7090	888	901	11186	1332	1336

Table 2: The total number of sentences after transforming the dataset into CSV format.

MAMS Dataset	ACSA			ATSA		
	Train	Validation	Test	Train	Validation	Test
Total unique Words	10458	2558	2640	13225	3249	3272
Longest Sentence	67	58	64	67	59	64
Shortest Sentence	4	5	5	4	5	5

Table 3: Dataset analysis after preprocessing.

review dataset.

The product links were gathered manually and stored in a JSON file. In order to make the process as automatic as possible, we fed the scraper in batches, depending on the number of reviews per phone, since there was a request limit over time. We manually selected 120 phones with a decent number of reviews and assigned an ID to each phone, with the main goal being that the number of reviews was at least 50. This yielded a total of 38,953 reviews. The scraper collected only the following data for each review:

- *Phone Model: The ID of one of the 120 chosen phones*
- *Index: A review ID to keep track of them*
- *Title: The review title*
- *Review: The review text body*
- *Star Rating: The grade the users gave, we saved it as integers from 0 to 100 but with increments of 20*
- *Likes: Total number of likes reviews had from other users.*

For training the models, we will use only the Title and Review. To gain insight into the rating, we can rely on the Star Rating. As for the number of likes, we haven't yet identified a clear use case, but it has been helpful in analyzing reactions to different types of reviews. For example, many Romanians tend to appreciate satirical or comedic reviews.

4.1.3 Analyse dataset

To ensure that the majority of our dataset was included in the Romanian dictionary, we scraped <https://ro.wiktionary.org/> [Wiktionary contributors, 2025] using Selenium. We then added an online dictionary containing some words that did not match. With all that data, another important step was to merge the two dictionaries into one. After compiling this

information, we copied all the data into another text file where all diacritics were replaced with their international letter equivalents. Finally, the dictionary was merged again, ignoring duplicates.

To evaluate the effectiveness of that dictionary in covering a dataset of Romanian product reviews, we calculate the coverage as the percentage of words in the dataset that are recognized by the dictionary. Formally, let W represent the set of words in the dataset, and D denote the dictionary, which may be either raw or lemmatized. The coverage (C) is defined as:

$$C = \frac{|W \cap D|}{|W|} \times 100 \quad (2)$$

The computation begins by preprocessing each review by lowering the text and removing the punctuation to standardize the input. Next, the text is tokenized into individual words. In order to improve dictionary matching, we optionally lemmatize the tokens using pre-trained Romanian NLP models from spaCy [Honnibal et al., 2020], including the small, medium, and large models. Lemmatization reduces words to their base forms, allowing recognition of multiple inflected variants in the dictionary.

After preprocessing and optional lemmatization, we count the occurrences of each unique word in the dataset. Each word is then checked against the dictionary to determine whether it is recognized or unrecognized. The coverage percentage is then calculated using the formula above, separately for raw and lemmatized words.

To visualize the dictionary coverage, we generate bar plots comparing the percentage of recognized words for raw and lemmatized datasets. Separate plots are created for each spaCy model (small, medium, and large) to illustrate how model size and lemmatization quality impact coverage.

We analysed the coverage plots to identify the most effective lemmatization approach for our dataset. When comparing the raw dictionary against the raw reviews, 70.4% from raw Reviews (the ones extracted from scraper) and 70.4% from the cleaned Dataset used for the model of the words were recognized, indicating a baseline coverage of the dataset. After applying lemmatization using the spaCy small model 7, coverage increased significantly: 88.3% of words from the lemmatized reviews and 88.0% words from lemmatized clean reviews matched entries in the lemmatized dictionary. Using the medium model 8 for lemmatization showed only a small improvement 89.9% of words from the lemmatized reviews and 89.5% words from lemmatized clean reviews matched entries in the lemmatized dictionary, meaning that the additional linguistic features of the medium model provided minimal benefit in this context. Interestingly, applying the large spaCy model 9 did not result in a different coverage over the medium model, indicating that for this specific dataset, the small model offers the perfect balance between coverage and efficiency. These results suggest that lemmatization substantially improves dictionary coverage, while increasing model size doesn't improve the coverage that much.

After analysing the data, 14 aspects were extracted: experience, price-quality, durability,

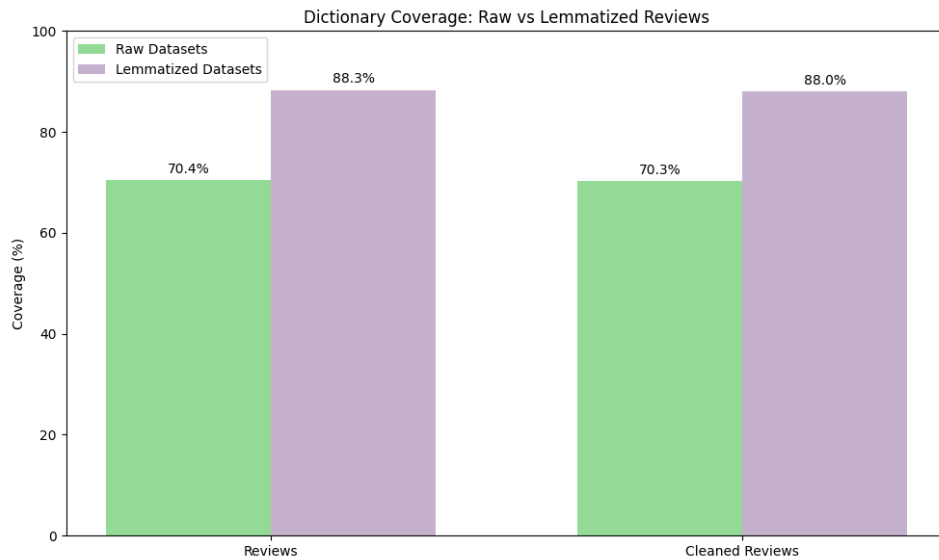


Figure 7: Word coverage of Raw Reviews vs. Raw Dictionary (left) and Lemmatized Reviews vs. Lemmatized Dictionary (right), using spaCy Small.

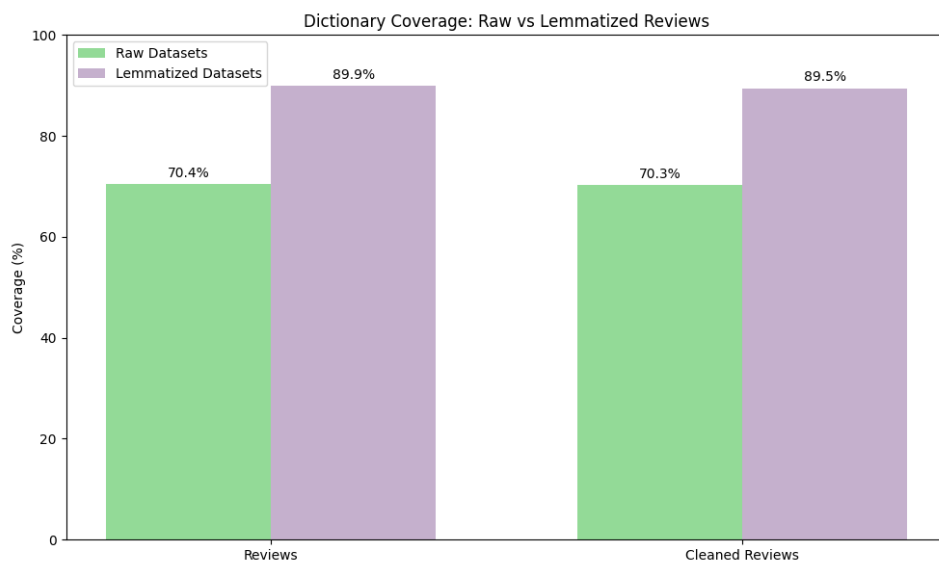


Figure 8: Word coverage of Raw Reviews vs. Raw Dictionary (left) and Lemmatized Reviews vs. Lemmatized Dictionary (right), using spaCy Medium.

camera, audio, battery, brand, design, quality, performance, accessories, compatibility, software and service, each one having 3 polarities: Positive, Negative and Neutral.

The way of expressing it is a little different depending on the brand, for example for Apple they are very focused on the brand (common expressions are; "Apple e Apple", "iPhone si atat", "calitate Apple") and on giving high marks even though the aspects are negative, some people do this for Samsung too, but not so much. For the other brands they are blunter, this is clearly observable in the mark.

It was also observed that over time people became less vocal about the appearance of

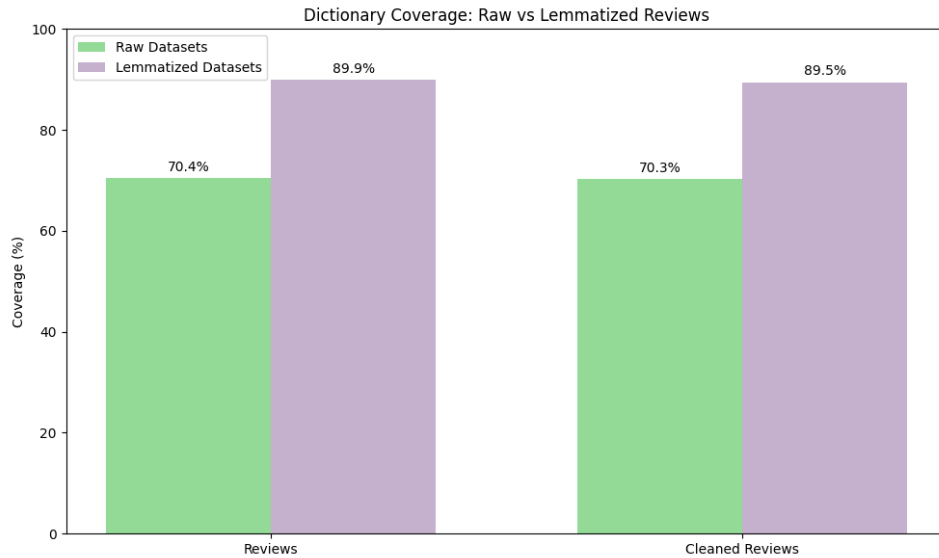


Figure 9: Word coverage of Raw Reviews vs. Raw Dictionary (left) and Lemmatized Reviews vs. Lemmatized Dictionary (right), using spaCy Large.

accessories, initially when they reduced the accessories received with the phone (e.g., charger, headphones etc.), a lot of negative reviews appeared, but over time, people accepted it as normal, very rarely mentioning the appearance for newer phones. Also over time, reviews became more minimalist, people wrote less.

The LLMs do not understand Romanian irony, for example, the quotation marks are often used to express the opposite meaning of the word for example the following review.

ID	Review	Manual Annotation	GPT-4.1	GPT-3.5
3440	Foarte multumit. Livrat ca de obicei, "prompt"! Despre telefon, este Iphone!	service: negative; brand: neutral; experience: positive	service: positive; brand: positive	service: positive; brand: neutral

Table 4: Comparison of manual annotation vs. GPT-4.1 and GPT-3.5 Turbo outputs.

Some reviews are quite funny, so even though the review is long, few aspects can be extracted. Examples: when the phone heats up, the phrase "va recomand numai în cazul în care vreți să vă încălziți mancarea" was used, for the monotony of the brand, phrases such as "aceasi marie" from the phrase "aceasi marie cu alta palarie" were used several times or "acesi ciorba reincalzita" was also encountered. To express that they gave quite a lot of money, they use phrases equivalent to "face mancare", "face ordine", "da cu aspiratorul".

The most common aspect is expectation, often being reviews that only express the fact that they are satisfied with the phone or they like it. Thus, the share of positive polarity is quite high, but since the dataset is built on real data.

Extremes such as maximum sentences, minimum sentences, maximum words and minimum

words were saved in a JSON file. Top used words and minimum used words were also saved as CSVs and are represented in Figure 10 is a simple plot of the top 10. Most used are mostly stopwords, for the least used, even though there are many that are addressed only once in the entire dataset, in the figure is a sample of them.

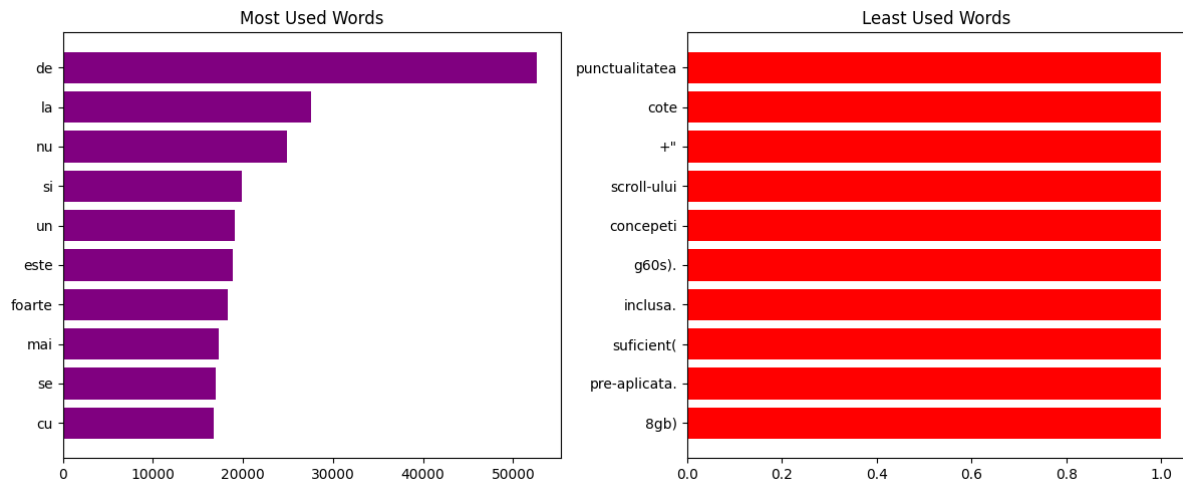


Figure 10: Top 10 most used words (left) and least used words (right).

The Figure 11 represent the word count per review, the most frequent words are in between 4 and 15 words per review, the average only being 40.9, as the number of words per review is rasing, the frequency is dropping. Another representation of this image would be how many character, including spaces, are per review, it can be observed that most people prefers using less characters to express their opinion on the phone category, the average would be 230.5 characters per review.

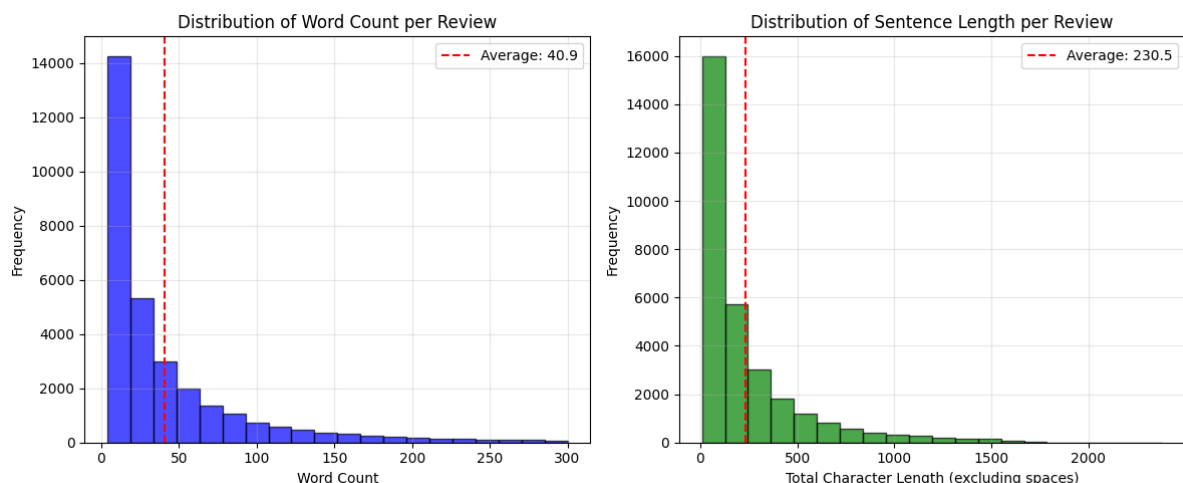


Figure 11: Distribution of the total words (left) and characters per review (right).

The Figure 12 is not surprising, but is the first one to start with a frequency bellow average, it is observed that the Romanian language doesn't contains verbose words, the plot reaching 0 frequency before 20 characters per word. This study says more about Romanian language than it says about the phone users.

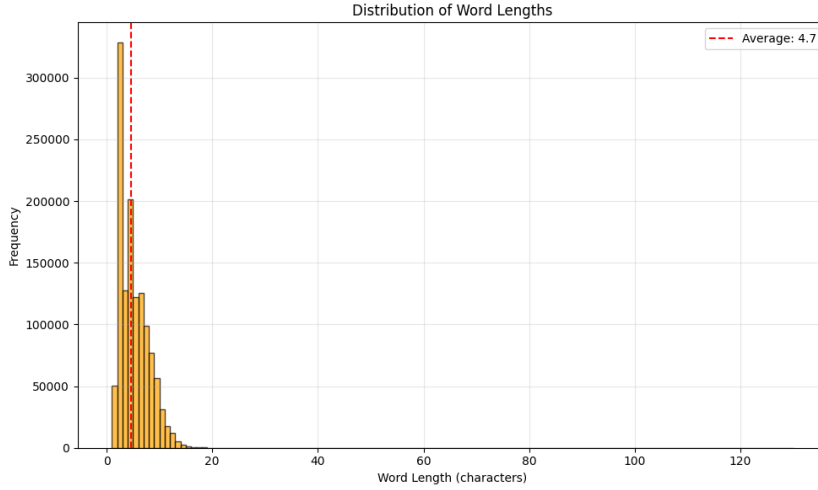


Figure 12: The lengths for each word in the entire dataset.

Nowadays, the most common form of communication is represented by emojis, some are universal in each country, others containing deeper meaning considering of the period of time the phone user is born and the region. In Figure 13, we represent top 10 most used emojis, it was observed that for the phones designed for seniors, the emojis were rarely present. Most common was like emoji, that is used more than 200 times in our dataset.

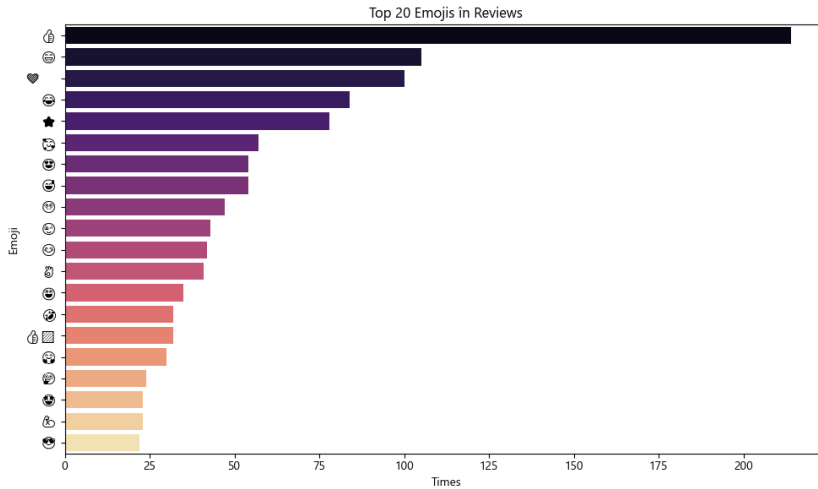


Figure 13: Top 20 most used emojis in the entire dataset.

Table 5 offers an overview of the RoR dataset used in our experiments, including the train, test, and combined sets. For each dataset, we report the total number of the reviews, the total word count across both the title and review text, and the number of aspect-level sentiments categorized as Neutral, Positive, and Negative.

From the data, the train set contains 27,630 reviews with a total of 1,201,316 words, of which 4,552 are neutral, 65,704 are positive, and 20,522 are negative aspect sentiments. The test set consists of 3,093 reviews totalling 111,747 words, with 827 neutral, 6,908 positive, and 1,743 negative aspect sentiments. The combined set (RoR) merges both train and test

sets, yielding 30,723 reviews and 1,313,063 words, with 5,379 neutral, 72,612 positive, and 22,265 negative aspect sentiments.

Table 5 highlights the distribution of sentiment-labelled aspects across the datasets, showing that the majority of the extracted aspect sentiments are positive, followed by negative and neutral, offering an overall distribution of the sentiments.

Dataset	Reviews	Words	Neutral	Positive	Negative
Train	27,630	1,201,316	4,552	65,704	20,522
Test	3,093	111,747	827	6,908	1,743
Combined	30,723	1,313,063	5,379	72,612	22,265

Table 5: RoR ABSA datasets including review counts, total words, and overall sentiment distribution.

4.1.4 Clean dataset

For cleaning, we’ve removed URLs, normalized whitespace, and applied word replacements for typos, abbreviations and informal text. We kept only reviews of 4 and 300 words. In the case where the title had more words than the review, they were swapped. Duplicate reviews were also removed. This yielded a total of 30,723 reviews.

P_ID	R_ID	Title	Review	Stars	Likes
42	23479	Super	A durat puțin până s-a obișnuit cu el persoana pentru care l-am cumpărat, dar acum e bun.	100	1
79	33300	Super	A durat puțin până s-a obișnuit cu el persoana pentru care l-am cumpărat, dar acum e bun.	100	3
65	30617	Nu recomand	Bateria nu ține mai mult de o zi.	20	2
32	21106	Excelent	Bateria nu ține mai mult de o zi.	100	0
114	38261	Recomand	Culoarea este superbă. A ajuns azi. Sper ca bateria să țină.	100	10
25	16836	Recomand	Culoarea este superbă. A ajuns azi. Sper ca bateria să țină.	100	1

Table 6: Examples of Review Duplicates, where P_ID represents the phone ID (from a specific phone), the R_ID represents the review ID for tracking, *Title* is written by the user (not they are different in some cases), *Review* text body (this is checked for for duplication), *Stars* which is the grading the user given to the phone, and the *Likes* received by other users.

Typo / abbreviation	Correct form
strano	straniu
memoie	memorie
descrie	descriere
repartie	reparatie
apid	rapid
telfon	telefon
oroduse	produse
exteem	extrem
funționează	funcționează
fuctioneaza	functioneaza
flima	filma
imd	imediat

Typo / abbreviation	Correct form
amintampinat	am intampinat
exep lu	exemplu
csf	ce sa faci
smartfone	smartphone
aman2	amandoua
aapărut	a apărut
ablocat	a blocat
abosult	absolut
about	despre
tel	telefon
ptr	pentru
pt	pentru
abroape	aproape

Table 7: Examples from the word replacements for typos and abbreviations.

4.1.5 Manual annotator tool for ABSA

Since we gathered a large amount of data, classical annotation was exhausting. To address this challenge, we created an application with Flask to make the process easier. The application includes all 14 aspects (experience, price_quality, durability, camera, audio, battery, brand, design, quality, performance, accessories, compatibility, software, and service) as well as the 3 polarities (positive, negative, and neutral).

You can select a CSV file, and the application will display the next unannotated review from that file. This approach proved to be much more effective, and productivity increased significantly. The functionality is minimalist: you can save by pressing the Next or Previous button, and to add another aspect, there is an "Add Aspect" button.

After the annotation of the test dataset, we decided to analyse the distribution of the sentiments and aspects. Figure 15 (left) represents the distribution of the sentiments for each aspect we have, noticeable would be the fact that the experience aspect is the most used in our test set, and the compatibility could be merged into performance since the representation in the set is barely noticeable. Figure 15 (right) is based on the overall sentiments, is observed that the distribution is unbalanced, 72,9% being positive, then is followed by negative with 18.4%, remaining only 8.7% for the neutral.

4.1.6 Mixed Dataset

After the test was manually annotated, a new pattern on the analysing part was discovered. Since we didn't label the dataset as single aspect or multiple aspects per review, we discovered

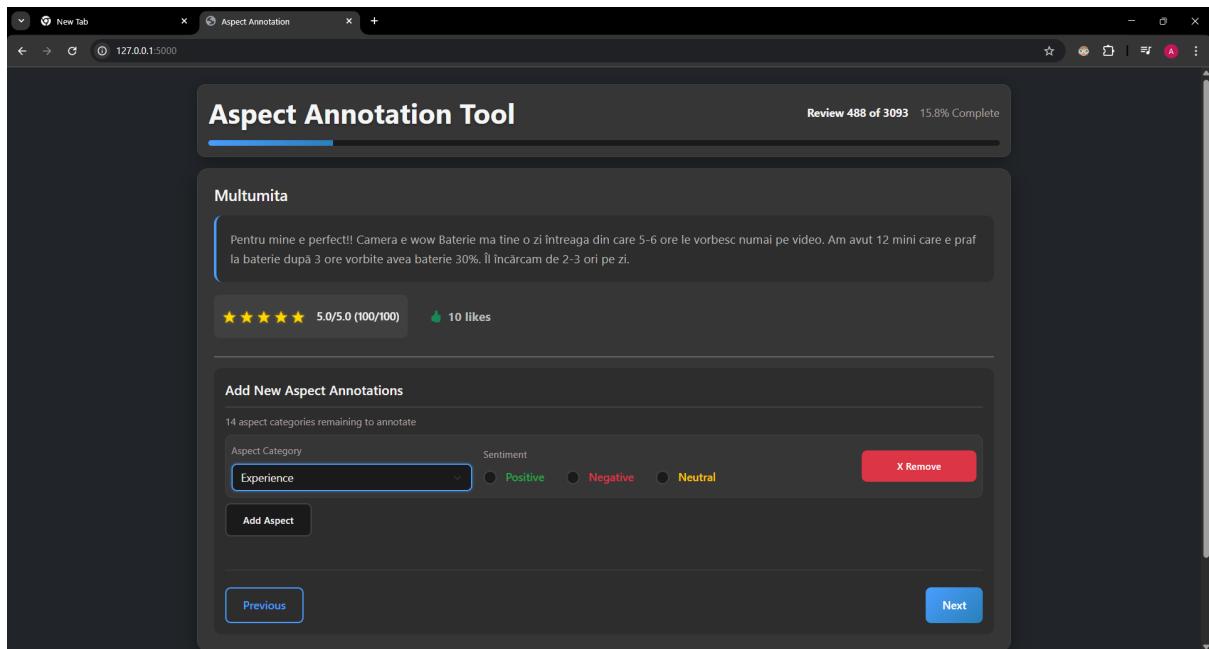


Figure 14: The aspect annotation tool interface showing a Romanian product review with its data, sentiment classification options and progress tracking.

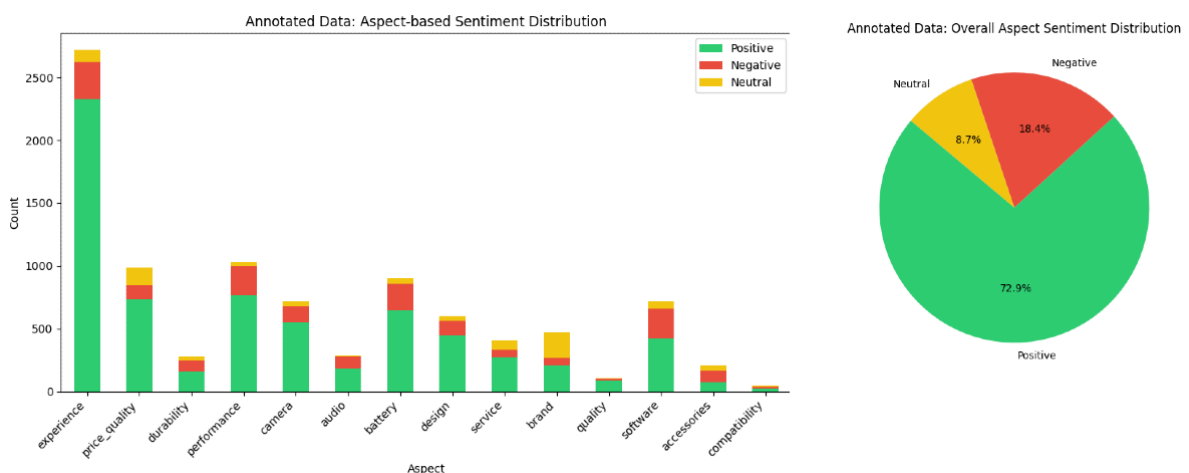


Figure 15: Histogram of the aspects manually annotated and Sentiment distribution over the annotated aspect TEST data

that it is actually both, resulting in a Mixed-Aspects dataset.

Figure 16 represents the distribution of reviews by the number of annotated aspects for the test, train, and combined sets. In the test set, 19.5% of reviews contain a single aspect, while 80.5% capture multiple aspects with potentially different sentiments. The train set exhibits 16.4% single-aspect and 83.6% multi-aspect reviews, and the combined dataset maintains a similar distribution (16.7% single-aspect, 83.3% multi-aspect).

These results highlight that the majority of reviews express mixed aspects and multi-sentiment information, which is critical for ABSA tasks. The consistent distribution across datasets

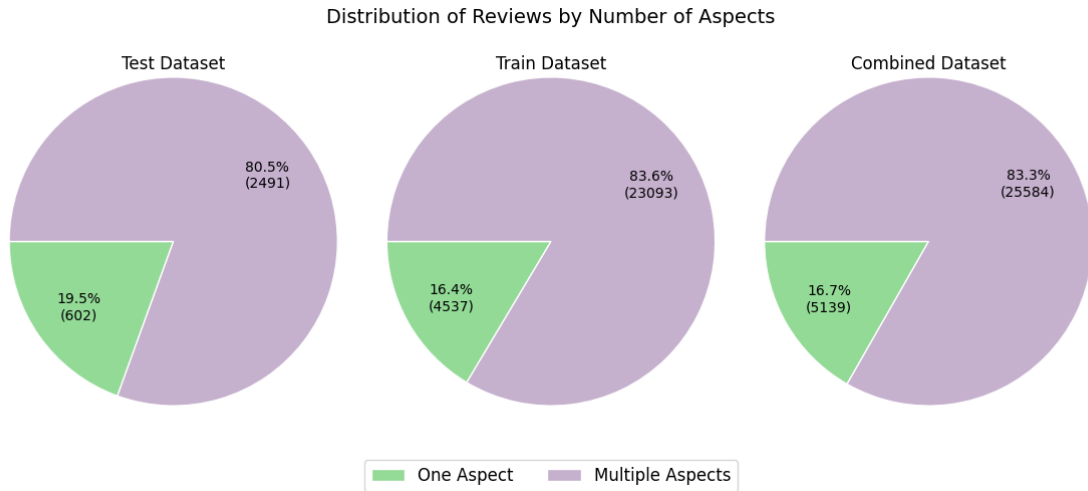


Figure 16: Distribution of reviews by aspect complexity across test (left), train (middle) and combined (right) datasets. Green represents the percentage of reviews with only one aspect and purple represents the percentage of reviews that have multiple aspects

indicates that both test and train sets represent similar complexity in aspect coverage, ensuring that models trained on these datasets are exposed to realistic multi-faceted user opinions and can generalize to multi-aspect sentiment extraction.

4.1.7 Automated annotation for train data with LLMs

Manual annotation is time-consuming, so we developed an automated annotation tool that uses Large Language Models inference to generate annotations for each review. The LLM takes detailed instructions as a prompt along with a few examples manually annotated reviews showing aspects and sentiments (few-shot learning). To improve efficiency, we batched the reviews into groups of 10 reviews per prompt, resulting in 3,074 total prompts per model. With annotated training data and a good human labelled benchmark dataset, we can prepare the training for the RoR transformer model. This approach follows a process similar to the knowledge distillation architecture [Hinton et al., 2015], where a larger, general-purpose model creates training data for a smaller, specialized model.

The prompt design follows a structured approach to ensure consistent and accurate aspect-based sentiment analysis annotation. It specifies the assistant's specialization in ABSA for Romanian customer reviews. It then provides aspect definitions for each predefined aspect categories with detailed explanations of each aspect scope to prevent category overlap and improve classification. The prompt incorporates several constraints such as mandatory JSON-only output format, aspect matching from the predefined list and, strict sentiment polarity options. To handle edge cases, the prompt includes fallback rules that default to experience with neutral sentiment when reviews are ambiguous. The user content section implements few-shot learning by providing five annotated examples that demonstrate the expected input and output format, showing how simple and complex reviews with multiple aspects should look

like. Each example includes the review ID for tracking, title, content, and the JSON structure with aspect-sentiment pairs.

Initially, we used GPT 3.5 Turbo [OpenAI, 2023] until the prompt and good performing parameters were found since it was cheap yet a capable model. We used a temperature setting of 0.3 to balance creativity with deterministic outputs, hoping for consistent annotation across similar reviews. The maximum token limit was set to 4,096 to accommodate batch processing of 10 reviews while allowing sufficient space for detailed review texts and JSON responses. The timeout was set to 60 seconds per API call to handle potential network latency while preventing blocking. We also used GPT 4.1 [OpenAI, 2025] since its a stronger and still affordable model.

The results were promising, but the models made a few mistakes regardless of the given prompt. There were mismatches between the title and review text in given input versus the generated output, this was either typo corrections or minor hallucinations. There was also sentiment mismatch, with the addition of a "Mixed" sentiment category since some users discussed about multiple products in the same prompt rather than focusing on the product in question (e.g., commenting about the battery being good for this phone but bad for their old one). Aspect mismatch was the most common mistake of the LLM approach, with many synonyms appearing for the given aspects.

To handle these errors and create consolidated training data that reflects real data from actual users, whom make mistakes and don't necessary have clear typing, we kept the original reviews, this would not be necessary if we can give preprocess the reviews at train and inference time to have those corrections applied. The sentiment mismatch was mapped from "mixed" to "neutral". The aspect mapping required more human intervention since there as a total of 99 unique added aspects. Some examples were "recommend" to "experience", "water resist" to "durability", "user friendly" to "design" and so on. This mapping generated some overlapping aspects (e.g., the memory being neutral, while performance was also present and its positive). To handle this, they were normalized by gibing highest priority to the current valid aspect rather than to the mismatch. If the mismatched mapping didn't exist in the current aspect list, then it was added to it.

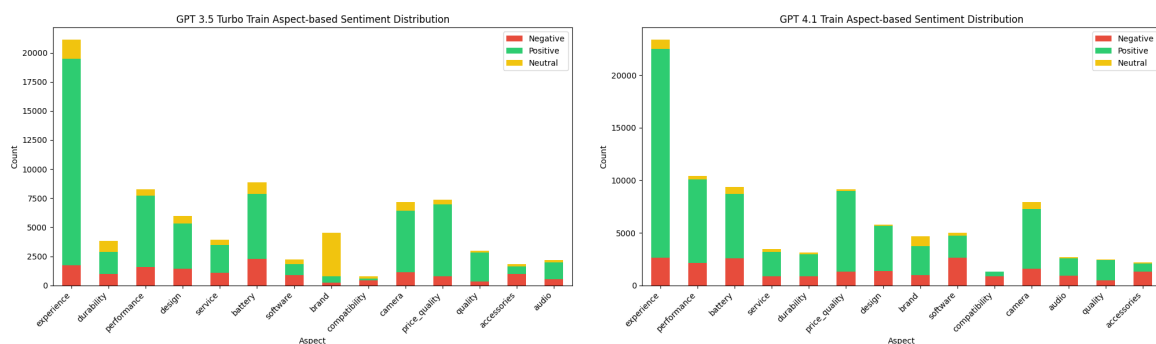
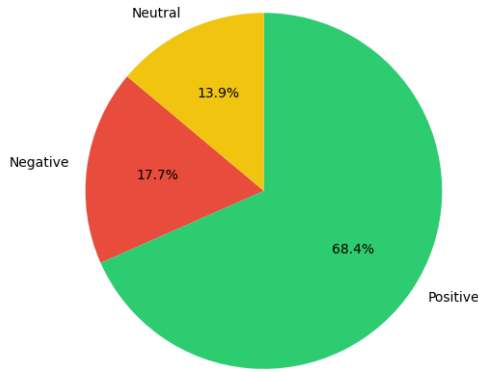


Figure 17: Histogram of the generated aspects over the train dataset with GPT 3.5 Turbo (left) and GPT 4.1 (right)

GPT 3.5 Turbo Train Overall Generated Sentiment Distribution



GPT 4.1 Train Overall Generated Sentiment Distribution

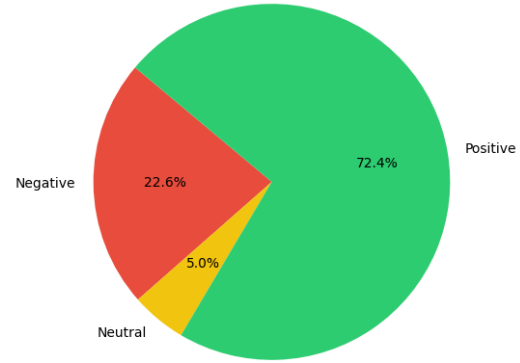


Figure 18: Generated sentiment polarity distribution over train data with GPT 3.5 Turbo (left) and GPT 4.1 (right)

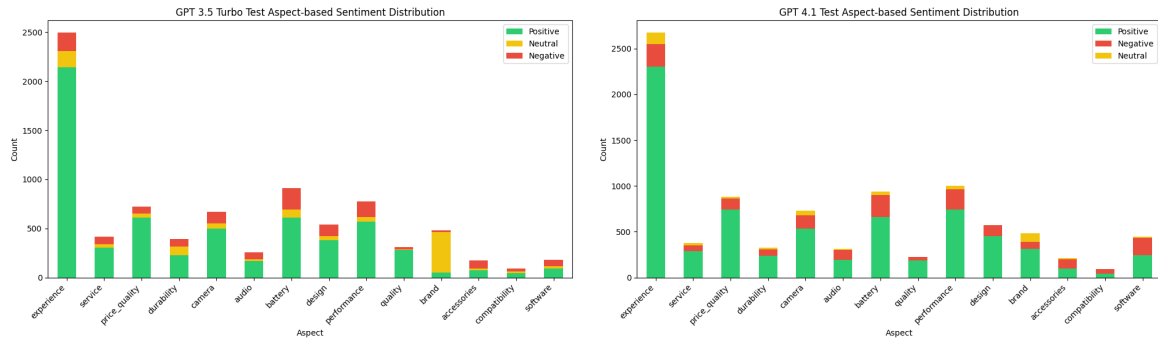


Figure 19: Histogram of the generated aspects over the test dataset with GPT 3.5 Turbo (left) and GPT 4.1 (right)

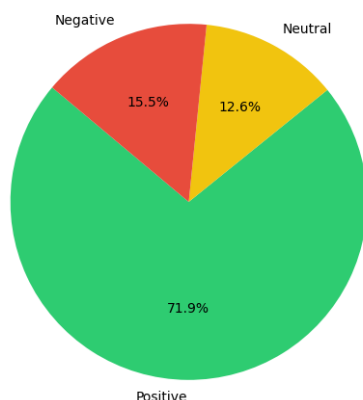
4.2 Modules implementation

This section presents the details of the implementation of the key system components: the preprocessing pipeline for Romanian text normalization for training, the data label processing for multi-label aspect-based sentiment analysis, and the hierarchical multi-task neural network architecture [O'shea and Nash, 2015].

4.2.1 Preprocessing

The preprocessing pipeline centres around a modular preprocessing function that uses multiple normalization techniques while preserving Romanian linguistic characteristics. The pipeline begins with HTML decoding, converting encoded characters (e.g., & to &) back to their original form. HTML tags are also removed using a regular expression. URLs are replaced with

GPT 3.5 Turbo Test Overall Generated Sentiment Distribution



GPT 4.1 Test Overall Generated Sentiment Distribution

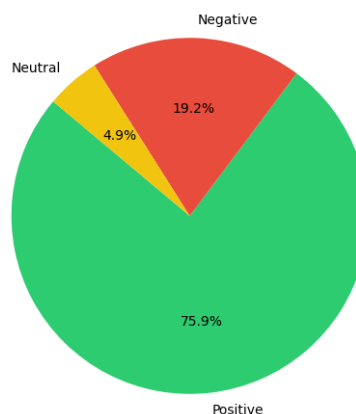


Figure 20: Generated sentiment polarity distribution over test data with GPT 3.5 Turbo (left) and GPT 4.1 (right)

standardized `<URL>` token using pattern matching for both `http` and `www` prefixes. Similarly email addresses are normalized to `<EMAIL>` tokens and long numerical sequences such as phone numbers or identification codes are replaced with `<NUM>` tokens.

Another preprocessing component addresses Romanian diacritical inconsistencies. The pipeline converts legacy encodings (e.g., "ț", "ț" or "ș", "ș"), correcting common encoding errors (e.g., "î", "î" or "ă", "ă").

An emoji to Romanian preprocessing module was developed to handle emoji conversion to appropriate Romanian text. This system utilizes a custom emoji dictionary mapping Unicode emoji characters to their Romanian textual equivalents. The implementation includes a fallback mechanism using Python emoji library when custom mappings are unavailable. The emoji replacement process identifies emoji positions and performs replacements from right to left to maintain correct text positioning.

Currency symbols are converted to their textual representations (e.g., \$ to "dollar") and mathematical symbols are normalized to improve tokenizer performance and understanding. Star rating are converted from textual character form, degree symbols are expanded (e.g., 32°C to "32 grade Celsius"). Punctuation marks were also normalized, converting different dash types to standard hyphens and normalizing quotation marks to consistent double quotes.

4.2.2 Data label processing

Romanian Review Dataset class implements a data loading and preprocessing system, designed for multi-label aspect-based sentiment analysis. The dataset handles the 14 distinct aspects and three sentiment categories. It supports both JSON and CSV input formats, with specialized

parsing methods for each format.

Aspects are encoded using binary vectors where each position corresponds to a specific aspect, allowing for multi-aspect assignment per review. Sentiments are represented as one-hot encoded vectors for each aspect, creating a 3D tensor structure (i.e., $aspects \times sentiments$) that captures the sentiment associated with each identified aspect.

Since review titles sometimes contain useful information, the titles were concatenated with the review text to capture more information for our model. The dataset implements an optional caching mechanism that pre-tokenizes all texts during initialization. This approach was implemented to reduce training time.

4.2.3 Model training

The RoR model architecture follows a hierarchical multi-task learning approach where aspect detection serves backbone for the sentiment classification. This design reflects the natural ordering of the ABSA task, the aspects must first be identified before their associated sentiments can be accurately classified. The architecture consists of three main components: a transformer-based feature extractor, an aspect classification head, and an aspect-conditioned sentiment classification head.

The model uses pre-trained Romanian transformer models as the foundation for feature extraction, specifically supported models from the Hugging Face ecosystem such as RoBERTa and BERT Romanian cased V1. The transformer component processes input text sequences to extract features from the special [CLS] token, which provides sentence-level representation. The model includes a freeze layers method that enables selective freezing of transformer layers.

The aspect classifier is implemented as a feed-forward neural network that predicts multiple aspects simultaneously. Its architecture includes a configurable number of layers, with optional batch normalization and dropout for regularization. The classification head projects the transformer's hidden dimension into a configurable hidden size-set to 256 in our case-followed by batch normalization when applied. The network then continues with a ReLU activation function, dropout regularization and concludes with a final fully-connected layer producing logits for each of the 14 aspects. This produces binary predictions for each aspect, enabling the detection of multiple aspects within a single review.

The sentiment classifier receives concatenated features composing the original transformer output and the sigmoid-activated aspect probabilities. This approach provides explicit aspect context for sentiment prediction and creates a dependency between the two tasks that reflect their natural relationship. This classifier produces logits for all aspect-sentiment combinations (i.e., $14aspects \times 3sentiments$, 42 outputs).

For training the losses from aspects and sentiments are combined, reflecting the multi-task nature of our ABSA problem. The loss function is designed to handle the hierarchical

relationship between tasks. Binary Cross-Entropy loss is applied to aspect predictions, treating each aspect as an independent binary classification problem. Then Cross-Entropy loss is applied only to aspects that are actually present in the ground truth labels, implementing a selective loss computation. This masking approach ensures that the model is not penalized for sentiment predictions on aspects that are not present in the review.

The loss functions use an optional class weighting for sentiment categories to address potential imbalances in training data since we have many positive reviews but not so many neutral ones. The final loss combines aspect and sentiment components with configurable weights, α and β mentioned in Equation 1, this allows for specific task emphasis during training. The higher weight for sentiment loss, for example, the more will the model focus on that task.

5 EXPERIMENTAL RESULTS

In this chapter, we present the experimental evaluation of our knowledge distillation approach for Romanian aspect-based sentiment analysis. We detail the training framework, experimental setup and, present results comparing fine-tuned Romanian transformer models trained on synthetic datasets generated by LLMs.

5.1 Experimental setup

The training framework implements F1 score metric tracking over both aspect and sentiment individually and early stopping capabilities. The training loop uses the AdamW [Loshchilov and Hutter, 2017] optimizer with configurable learning rate with default $2e - 5$ and implements a linear learning rate scheduling with warmup, the scheduler is called at the end of each batch.

The validation dataset represents 20% of the training data, and its metrics are used to evaluate the quality during training. Both training and validation phases compute losses such as the total loss, and component losses (i.e., aspect and sentiment) and F1 scores for both tasks. F1 scores are calculated using weighted-averaging to provide balanced performance for all classes. Sentiment metrics are calculated only when relevant aspects are present, when no aspects are detected in a batch, sentiment F1 score defaults to zero. To prevent overfitting the early stopping was implemented to monitor validation loss with a default patience of 2 epochs and a minimum improvement threshold default of 0.001. The system automatically saves the best model weights according to validation performance.

The training framework uses a configuration class and file that encapsulates all experimental parameters, ensuring reproducibility and experimentation. Post-training it performs an evaluation against our manually annotated dataset and compares with its teacher model through numerical and visual evaluation.

5.2 Results

We conducted experiments mainly focused on training Romanian transformer architectures as our foundation models RoBERT and BERT Romanian cased V1 on the generated datasets. Each model was trained on two different automatically generated datasets, one created using GPT 3.5 Turbo and another using GPT 4.1 annotations, resulting in four distinct fine-tuned models.

The experimental results demonstrate that our knowledge distillation approach successfully transfers the reasoning capabilities of large language models to smaller, more efficient transformer architectures. Our fine-tuned Romanian transformer models, approximately 100M parameters each, achieve comparative results with their teachers, obtaining F1-scores of 0.788 (BERT Romanian V1) and 0.787 (RoBERT), compared to GPT 4.1 0.791 F1-score as shown in Table 8.

The quality of the training data provides crucial for model performance. Models trained on GPT 4.1 generated annotations consistently outperform their counterparts trained on GPT 3.5 Turbo on all performed experiments.

Model Type	Train Dataset	F1-Score
GPT 3.5 Turbo	No Train: Few-Shot	0.692
GPT4.1	No Train: Few-Shot	0.791
RoBERT	Generated GPT 3.5 Turbo	0.725
RoBERT	Generated GPT4.1	0.787
BERT Romanian V1	Generated GPT 3.5 Turbo	0.664
BERT Romanian V1	Generated GPT4.1	0.788

Table 8: Model performance comparison across different training datasets.

Interestingly, both Romanian transformer models achieve nearly identical performance when trained on higher quality data generated by GPT 4.1, suggesting that the quality of training annotation is more critical than specific choice between these Romanian language models.

These results demonstrate that achieving good performance through strategic use of synthetic data is possible while using less resources, making our approach more suitable for production deployment. The main advantages reduced inference costs, faster response times suitable for real-time applications, lower memory consumption and eliminating the API dependencies for privacy concerns.

6 CONCLUSIONS

This research successfully created the Romanian Review (RoR) dataset, one of the first, to our knowledge, Mixed-Aspect Multi-Sentiment ABSA dataset for Romanian language. The dataset contains a total of 30,723 cleaned reviews with 14 aspects and 3 polarities and it demonstrates that 83.3% of the real-world reviews contain multiple aspects with varying sentiments, addressing a critical gap in Romanian NLP resources.

Our knowledge distillation approach achieved competitive performance, with fine-tuned Romanian transformer models (BERT Romanian V1: $F1 = 0.788$, RoBERT: $F1 = 0.787$) nearly matching their GPT-4.1 teacher model ($F1 = 0.791$) while requiring significantly fewer computational resources. This demonstrates the viability of transferring complex ABSA capabilities to smaller models.

The RoR dataset addresses limitations in existing Romanian resources, which were previously restricted to single-aspect reviews or binary sentiment classification. Our dictionary coverage analysis revealed a 88.3% word recognition with lemmatization, confirming the dataset linguistic validity. The hierarchical multi-task learning architecture effectively handles the complexity of multi-aspect sentiment analysis in Romanian text.

We developed several key components such as an automated annotation pipeline using LLMs with few-shot learning, a web-based manual annotation tool that significantly improved manual annotation efficiency, and a comprehensive preprocessing pipeline tailored for Romanian text normalization. The knowledge distillation framework successfully transferred reasoning capabilities from large language models to smaller transformer architectures.

The dataset is far from perfect, it shows sentiment imbalance (72.9% positive, 18.4% negative and 8.7% neutral for the manually annotated test set), reflecting real-world review distributions but also affecting model training. Domain specificity limits generalization beyond phone reviews. Romanian irony and sarcasm remain challenging for automated annotation, as demonstrated by LLM misinterpretations of quotation marks and cultural expressions.

For the future, we would like to explore expanding the dataset to additional product categories and implementing clause-level analysis to handle repetitive aspects with varying polarities. Long-term goals involve a better handling of Romanian linguistic nuances, exploring cross-lingual transfer learning, and creating tools for real-time review analysis that could assist Romanian customers in making informed purchasing decisions.

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